MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION (COMBINED TECHNICAL) FOR RECRUITMENT TO THE POSTS OF AGRICULTURE & ALLIED SERVICES, GOVERNMENT OF MIZORAM.

(NOVEMBER-2025)

AH&VETY PAPER-II

Time Allowed : 3 hours FM : 200

SECTION - A *(Multiple Choice questions)* (100 Marks)

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

Notes regarding Negative Marking :

(i) There will be penalty (negative marking) for every wrong answer given by a candidate in the MCQ type questions.

(ii) For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty

(iii) If a candidate gives more than one answer to one question, it will be treated as a wrong answer even if any one of the given answers happens to be correct and there will be same penalty as above for that question.

(iv) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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1. Eschar is developed in the skin of human which is the hallmark lesion of:

(a) Scrub typhus (b) Q-fever

(c) Black death (d) Black leg

2. The presentation of endogenous antigen by APC takes place through _____ activating _____ response.

(a) MHC-I, humoral mediated immune (b) MHC-I, CMI

(c) MHC-II, humoral (d) MHC-II, CMI

3. One among the listed is true for *Listeria.*

(a) Gram positive, can grow in 4°C (b) Gram positive, cannot grow in 4°C

(c) Gram negative, can grow in 4°C (d) Gram positive, cannot grow in 4°C

4. Cancer causing virus are:

(a) Papillomavirus (HPV), Hepatitis B and Bovine Ephemeral Fever Virus

(b) Papillomavirus (HPV), Hepatitis B (HBV) and Retrovirus

(c) Papillomavirus (HPV), Epstein-Barr Virus, Asfaviridae

(d) Papillomavirus (HPV), Hepatitis B, C & Bornaviridae

5. The following statement about Marek's disease is true:

(a) It is caused by avian herpesvirus (MDV) which induce B-cell lymphomas and immunosuppression.

(b) It is caused by avian adenovirus serotype 4 (FAdV) which induce B-cell lymphomas and immunosuppression.

(c) It is caused by avian herpesvirus (MDV) which induce T-cell lymphomas and immunosuppression.

(d) It is caused by avian adenovirus serotype 4 (FAdV) which induce T-cell lymphomas and immunosuppression.

6. *Mycobacterium bovis* is:

(a) Gram positive, contains high lipid layer in the cell wall, spore forming

(b) Gram positive, contains high lipid layer in the cell wall, non-spore forming

(c) Gram negative, contains low lipid layer in the cell wall, spore forming

(d) Gram positive, contains low lipid layer in the cell wall, non-spore forming

7. The component of vaccine against Newcastle in poultry is:

(a) Live virus attenuated vaccine- La Sota strain

(b) Killed virus vaccine- La Sota strain

(c) Live virus attenuated vaccine- S19 strain

(d) Killed virus vaccine- S19 strain

8. The choice of disinfectant during the outbreak of CSF are:

(a) Sodium hypochloride, Caustic Soda, Potassium Permanganate

(b) Sodium hypochloride and Caustic Soda

(c) Sodium hydroxide and Potassium Permanganate

(d) Sodium hypochloride and Potassium Permanganate

9. Select the most correlating pathogens and the disease they cause:

(a) Brooders pneumonia (*Aspergillus*), Ephemeral fever (Rhabdoviridae), Infectious coryza (*Haemophilus*), PRRS (*Arterivirus*)

(b) Brooders pneumonia (*Histoplasma*), Ephemeral fever (Rhabdoviridae), Infectious coryza (*Haemophilus*), PRRS (*Adenovirus*)

(c) Brooders pneumonia (*Aspergillus*) Ephemeral fever (Rhabdoviridae), Infectious coryza (*Birnavirus*), PRRS (*Arterivirus*)

(d) Brooders pneumonia (*Aspergillus*), Ephemeral fever (Flavivirus), Infectious coryza (*Retrovirus*), PRRS (*Parvovirus*)

10. Humoral type of immunity is stimulated when the antigen induced:

(a) Th1 type of immune response

(b) Th2 type of immune response

(c) Cytotoxic type of immune response

(d) All of the above

11. High titre of IgM is seen during serological examination that indicates:

(a) Primary infection or initial stage of infection

(b) Secondary infection or chronic stage of infection

(c) Tumor formation

(d) All of the above

12. The stain generally used for identifying fungus is:

(a) Giemsa's stain (b) Lactophonol

(c) Lactophenol cotton blue (d) Zeihl-Nelson stain

13. Lactose intolerance happen where:

(a) The small intestine does not produce enough lactase

(b) The liver does not produce enough lactase

(c) The small intestine does not produce enough lactocase

(d) The liver does not enough lactocase

14. The nature of ASF virus is:

(a) Double stranded which multiply in the cytoplasm of the infected cell

(b) Single stranded which multiply in the cytoplasm of the infected cell

(c) Retro virus which multiply in the cytoplasm of the infected cell

(d) It can transform in any form which makes the virus extremely lethal for pigs

15. A cattle suffers from pyrexia, enlarge lymph nodes and nodules in the skin, (2-5 cm in diameter), nodules in mucous membrane, nasal and ocular discharges is indicative of:

(a) Theileriosis (b) Lympy Skin Disease

(c) Bovine Ephemeral fever (d) Foot and Mouth Disease

16. The transmission of FMD virus may be completed through:

(a) Aerosol, milk, semen (b) Aerosol and fomites

(c) Aerosol, feed and water (d) All of the above

17. Reproductive problems is the most obvious clinical signs in the infection of pigs among the disease listed below:

(a) CSF (b) ASF

(c) PRRS (d) Swine Influenza

18. The causative agent of greasy pig disease is:

(a) *Erysipelothrix rhusiopathiae* (b) *Staphylococcus hyicus*

(c) *Streptococcus suis* (d) *Clostridium perfingens*

19. Under microscopic examination of stained blood smear, various sizes of RBCs are seen which is indicative of:

(a) Erythropoiesis (b) Hyperthyroidism

(c) Acute viral disease condition (d) All of the above

20. During urine analysis, the level of bilirubin is quantified for:

(a) Acidity (b) Kidney stone

(c) Liver disease (d) UTI

21. The normal pH of milk is _____.

(a) 4.6 (b) 5.6

(c) 6.6 (d) 7.6

22. Souring & gassiness of milk is caused by _____ group, which indicates the contamination of milk and its products.

(a) Lactic acid (b) *Bacillus lactis*

(c) Coliform (d) Streptococcus

23. Clonorchiosis is an example of :

(a) Metazoonosis subtype II (b) Metazoonosis subtype III

(c) Metazoonosis subtype IV (d) Cyclozoonosis

24. During slaughtering of pig, the captive bolt should be placed _____.

(a) on top of head (b) middle of forehead

(c) 2.5 centimeters above eye level (d) 13 mm running down the centre of forehead

25. _____ is the type of vaccination that is practiced both within and around the region where an outbreak had occurred.

(a) Ring vaccination (b) Strategic vaccination

(c) Barrier vaccination (d) Suppressive vaccination

26. The first step in disease investigation:

(a) Analytical epidemiology (b) Horizontal epidemiology

(c) Descriptive epidemiology (d) Predictive epidemiology

27. Koch's postulate is not applicable to which of the following:

(a) Ebola (b) Mastitis

(c) Collibacillosis (d) Salmonellosis

28. The animal species which is most susceptible for tetanus is _____.

(a) dog (b) horse

(c) cattle (d) sheep

29. Example of algal zoonosis:

(a) Scrub typhus (b) Actinomycosis

(c) Fascioliasis (d) Protothecosis

30. Recording of events occurring in a particular point of time is called _____.

(a) Prospective survey (b) Retrospective survey

(c) Cross-sectional survey (d) Longitudinal survey

31. Relate the following hormone in **direct** connection to GnRH:

(a) FSH & LH (b) FSH, LH & Progesterone

(c) FSH & Testosterone (d) FSH, LH & TSH

32. The final product of oxidative phosphorylation is:

(a) NADH (b) ATP

(c) FADH2 (d) All of the above

33. Deamination is the_____of amino group from/to amino acid which occurs in _____.

(a) release, kidney
(b) release, liver
(c) uptake, liver
(d) uptake, kidney

34. Hypoglycemia in piglets is due to:

(a) Inefficient glycolysis
(b) Inefficient TCA cycle
(c) Insufficient colostrum and milk intake
(d) Inefficient gluconeogenesis

35. In nature pH level ranges from:

(a) 0-14
(b) 1-12
(c) 1-10
(d) 2-10

36. During skeletal muscle contraction, _____ binds to troponin.

(a) calcium
(b) magnesium
(c) sodium
(d) potassium

37. In the blood, oxygen binds to hemoglobin of the red blood cells (RBCs) to form _____ and is carried to the body's tissues.

(a) 2-Haemoxyglobin
(b) Oxyhaemoglobin
(c) Beta-haemoxyglobin
(d) Oxyglobinogen

38. The release of egg from ovary during oestrus/menstrual is stimulated by_____.

(a) Surge of LH level
(b) Dip in the LH level
(c) Surge in the FSH
(d) Dip in the FSH

39. RNA is more labile to hydrolysis than DNA because of the presence of _____.

(a) 2'-hydroxyl group (2'-OH) on its ribose sugar in RNA
(b) 2'-hydroxyl group (2'-OH) on its fructose sugar in RNA
(c) 5'-hydroxyl group (2'-OH) on its ribose sugar in RNA
(d) 5'-hydroxyl group (2'-OH) on its fructose sugar in RNA

40. The RBC of _____ is biconcave in nature.

(a) goat
(b) dog
(c) camel
(d) pig

41. Consumption of measly pork leads to an infection known as:

(a) Cysticercosis
(b) Taeniasis
(c) Hydatidosis
(d) All of the above

42. McLean County system is applicable only in *Ascaris suum* infection among the ascarid worm due to:

(a) Absence of somatic migration of the larvae
(b) Presence of somatic migration
(c) The large size stage two larva
(d) Thick albuminous covering of the egg

43. A dog suffering from high fever, anaemia/pale mucous membrane, ascites with tick infestation is indicative of _____ and treatment may be given with _____.

(a) *Neospora caninum,* Imidocarb

(b) *Trypanosoma evansi*, Diaminazine aceturate

(c) *Babesia gibsoni,* Imidocarb

(d) *Hepatozoon canis,* Diaminazine aceturate

44. Examination of the stool of immunocompromised person contains a short tail parasitic larva is an indicative of _____ and the confirmation is done by_____.

(a) *Strongyloides stercoralis*, Harada Mori technique

(b) *Ancylostoma duodenale*, Formol-ether technique

(c) *Schistosoma mansoni*, Mac Master Technique

(d) *Ascaris lumbricoides*, Salt floatation Technique

45. Cat is the only definitive infection for *Toxoplasma gondii* which is due to:

(a) Presence of delta-6-desaturase

(b) Absence of delta-6-desaturase

(c) Presence of alpha-6-desaturase

(d) Absence of alpha-6-desaturase

46. Which among the statements is true for *Sarcoptes?*

(a) The female parasite lay up to 3000 eggs (approx.) in a burrow and dies at the end of a tunnel

(b) The female parasite lay up to 30 eggs (approx.) in a burrow and dies at the end of a tunnel

(c) The female parasite lay up to 3000 eggs (approx.) in a burrow dig another tunnel

(d) The female parasite lay up to 30 eggs (approx.) in a burrow and dig another tunnel

47. Arrange the following in the chronological order:

(a) Adult, sporocyst, miracidium, redia, cercaria, metacercaria

(b) Adult, miracidium, redia, sporocyst, cercaria, metacercaria

(c) Adult, miracidium, sporocyst, redia, cercaria, metacercaria

(d) Adult, redia, sporocyst, miracidium, cercaria, metacercaria

48. The most appropriate relationship of the following intermediate host and a parasite is:

(a) Land snail, *Angiostrongylus cantonensis*

(b) Mosquito, *Spirocerca lupi*

(c) Water snail, *Dicrocoelium dentriticum*

(d) Fresh water fish, *Paragonimus westermani*

49. In apicomplexan parasite, recognition and penetration of a target cell is taken by the following organelles such as _____ & _____, respectively.

(a) Conoid, microneme

(b) Dense granule, rhoptry

(c) Sub-pellicular microtubule, conoid

(d) Microneme, Conoid

50. The etiology of alveolar hydatidosis is_____.

(a) *Echinococcus granulosus*

(b) *Echinococcus multilocularis*

(c) *Echinococcus vogeli*

(d) *E. oligarthrus*

SECTION - B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it.
*Attempt **any 5 (five)** questions.*

1. Answer the following:

(a) Describe various stages of estrus cycle; ovulation; fertilization and pregnancy in farm animals. **(10)**

(b) What is recombinant DNA? How is recombinant DNA construct generated? Mention the utility of recombinant DNA. **(10)**

2. Enlist *Eimeria* of chicken, write the pathogenesis of hemorrhagic and non-hemorrhagic coccidiosis in chicken, their clinical manifestation, post-mortem lesions and their control strategies. **(20)**

3. Answer the following:

(a) What is cell injury? Write different types of cell injury. **(10)**

(b) Write different types of skin diseases in animals due to various atiological agents. **(10)**

4. What is rabies? What is the etiology, transmission pattern, pathogenesis, clinical manifestation in man and in animals and control strategies of rabies? **(20)**

5. Answer the following:

(a) What is myasis in animal? **(5)**

(b) Write various generations of vaccines with example. **(5)**

(c) Differentiate Mareks' disease and Lymphoid Leukosis. **(5)**

(d) Write the protocol and reagents of PCR reaction. **(5)**

6. Answer the following:

(a) Write the hormones of various endocrine organs, their mode of actions, and the manifestation of imbalance, if any. **(10)**

(b) Write the mechanism of urine formation mentioning the significance of each of the structure nephron. **(10)**

7. Answer the following:

(a) Scabies in pigs, aetiology, the pathogenesis, clinical manifestation and diagnosis. **(10)**

(b) Parvo virus infection in dogs, the pathogenesis, clinical manifestation and treatment and control **(10)**

8. Answer the following:

(a) What is clean milk? How can clean milk production be achieved? **(10)**

(b) Define HAACP and explain the principles of HAACP. **(10)**

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